

Stony Brook University
Alejandro Melo Ponce

Spring 2018

Midterm 1

Take-home exam

Solutions

These are suggested solutions for the take-home exam. The answers appear immediately after each question with the respective R code.

First of all, when creating your R script, start by cleaning the working environment, setting up your working directory and loading some useful packages.

```
rm(list=ls())
#To clean the environment

setwd("~/Dropbox/ECO-321-Spring_2018/Midterms/Take-home_M1")
#set to your own working directory

#some packages we will use
library(foreign)
library(sandwich)
library(lmtest)
```

1. The data set in CEOSAL2 contains information on chief executive officers for U.S. corporations. The variable *salary* is annual compensation, in thousands of dollars, and *ceoten* is prior number of years as company CEO.

Start by loading the data set for this question, CEOSAL2.

```
load("ceosal2.RData")
```

- (a) Find the average salary and the average tenure in the sample.

The average salary is \$865.8644 and the average tenure is 7.95 years.

```
cat("The average salary is_", mean(data$salary), "\n")
cat("The average tenure is", mean(data$ceoten))
```

- (b) How many CEOs are in their first year as CEO (that is, *ceoten* = 0)? What is the longest tenure as CEO?

The number of CEOs who are in their first year is 5 and the longest tenure as a CEO is 37 years.

```
length(which(data$ceoten == 0))
max(data$ceoten)
```

- (c) Estimate the simple regression model

$$\log(\text{salary}) = \beta_0 + \beta_1 \text{ceoten} + u$$

and report your results in the usual form. What is the (approximate) predicted percentage increase in salary given one more year as a CEO?

The estimated model is

$$\widehat{\log(\text{salary})} = 6.5054 + \frac{.0097}{(0.00697)} \text{ ceoten}, R^2 = 0.01316, SER = 0.6038.$$

From the results, we can see that the coefficient of *ceoten* is $0.009724 \approx 0.01$. Thus, in prediction, the salary of CEO would increase by nearly one percentage point with an extra year of tenure.

```
#The regression object
regression_ceosal<-lm(log(salary)~ceoten,data=data)
#Summary of the previous estimation (NOTICE: This command reports
#homoskedasticity only errors,
#we will fix this with the next command)
summary(regression_ceosal)
#The command that outputs the estimation result with the robust
#standard errors.
coeftest(regression_ceosal, vcov = vcovHC(regression_ceosal, "HC1"))
```

2. Use the data in WAGE2 to estimate a simple regression explaining monthly salary (*wage*) in terms of IQ (*IQ*).

Start by cleaning the environment and loading the WAGE2 data set.

```
rm(list=ls())
#To clean the environment again

#load the wag2 data set
load("wage2.RData")
```

- (a) Find the average salary and average IQ in the sample. What is the sample standard deviation of IQ? (IQ scores are standardized so that the average in the population is 100 with a standard deviation equal to 15.)

The average wage is \$957.95 and the average IQ is 101.28 points.

```
cat("The average wage is_", mean(data$wage), "\n")
cat("The average IQ is", mean(data$IQ))
```

- (b) Estimate a simple regression model where a one-point increase in *IQ* changes *wage* by a constant dollar amount. Use this model to find the predicted increase in wage for an increase in *IQ* of 15 points. Does *IQ* explain most of the variation in *wage*?

The model that we will estimate is:

$$\text{wage} = \beta_0 + \beta_1 \text{IQ} + u,$$

which implies that $\frac{\partial \text{wage}}{\partial \text{IQ}} = \beta_1$, i.e. a one point increase in IQ would imply a constant β_1 increase in wage. The estimated model is

$$\widehat{\text{wage}} = 116.99 + \frac{8.3}{(0.8364)} \text{IQ}, R^2 = 0.09554, SER = 384.8.$$

The estimated change in wage given an increase in 15 IQ points is $\Delta \text{wage} \approx 8.3 \times 15 = \124.546 . Also, the R^2 tells us that the model explains approximately 9.5% of the variance of monthly salaries.

```
#The regression object
regression_wage<-lm(wage~IQ,data=data)
#Summary without robust errors.
summary(regression_wage)
#summary with robust errors
coeftest(regression_wage, vcov = vcovHC(regression_wage, "HC1"))
#extracting the coefficient on IQ and estimating the approximate
#change given a 15 point increase in IQ.
coef(summary(regression_wage))["IQ", "Estimate"]*15
```

- (c) Consider the following statement: *IQ has no impact on monthly salary*. How would you test such a statement? Propose a hypothesis test that allows to evaluate the previous statement. Perform the test at the 5% level. **Be sure to use robust standard errors.**

The statement is equivalent to the standard hypothesis test: $H_0 : \beta_1 = 0$ v.s. $H_a : \beta_1 \neq 0$. We can immediately test this hypothesis from the previous results. The t -statistic using robust errors is $t = 9.927203$ and the p -value for the respective two sided 5% test is 3.794935×10^{-22} . The critical value for a 5% two-sided test is 1.96. Since the t -statistic is greater than this value we reject the null. Alternatively, since the p -value is considerably smaller than 0.05, we also reach the same conclusion. In simple words, the results of this regression reject at the 5% level the statement that IQ has no impact on monthly salary.

```
#The results for the standard hypothesis test can be seen
#from the summary of the regression with robust errors.
coeftest(regression_wage, vcov = vcovHC(regression_wage, "HC1"))
```

3. For this problem, we will use the concept of *elasticity*. Suppose that we have a function $z = f(w)$. The elasticity is the percentage change in z given a percentage change in w . Then the elasticity of f at point w is given by

$$\epsilon_f(w) = \frac{\frac{\Delta z}{z}}{\frac{\Delta w}{w}} = \frac{\Delta z}{\Delta w} \times \frac{w}{z} = \frac{\Delta f(w)}{\Delta w} \times \frac{w}{f(w)}.$$

Notice that if we let the change in w be small, i.e. $\Delta w \rightarrow 0$ then we get that

$$\epsilon_f(w) = f'(w) \times \frac{w}{f(w)},$$

Now, define the variables $Z = \ln(z)$ and $W = \ln(w)$. Then we have that $\exp(Z) = f(\exp(W))$, so that $Z = \ln(f(\exp(W)))$. Thus, using the chain rule and other rules of differentiation, we have that

$$\frac{dZ}{dW} = \frac{d \ln(z)}{d \ln(w)} = \frac{1}{f(\exp(W))} \times f'(\exp(W)) \times \exp(W) = f'(w) \times \frac{w}{f(w)} = \epsilon_f(w).$$

Why is this useful? Consider the linear model $y = \beta_0 + \beta_1 x + u$. The interpretation of β_1 is that it is the marginal change in y given a one unit increase in x , i.e. the slope. Since the model is linear we have that $\frac{dy}{dx} = \beta_1$. But what if we write the model in “logs”? We would write $\ln(y) = \gamma_0 + \gamma_1 \ln(x) + v$. Then the interpretation of γ_1 is different, it is no longer the marginal change, but the *percentual change in y given a percentual change in x* , i.e. now γ_1 is an elasticity.

Now, after this introduction, consider the following problem. For the population of firms in the chemical industry, let *rd* denote annual expenditures on research and development, and let *sales* denote annual sales (both are in millions of dollars).

- (a) Write down a model (not an estimated equation) that implies a constant elasticity between rd and $sales$. Which parameter is the elasticity? The following model specification

$$\log(rd) = \gamma_0 + \gamma_1 \log(sales) + v$$

implies a constant elasticity between rd and $sales$, since

$$\frac{\frac{\Delta rd}{rd}}{\frac{\Delta sales}{sales}} = \frac{\partial \log(rd)}{\partial \log(sales)} = \gamma_1,$$

where the elasticity is of course γ_1 .

- (b) Now, estimate the model using the data in RDCHEM. Write out the estimated equation in the usual form. What is the estimated elasticity of rd with respect to $sales$? Explain in words what this elasticity means.

After estimating the model, we get the following estimated equation:

$$\widehat{\log(rd)} = \underset{(0.421)}{-4.1047} + \underset{(0.0573)}{1.0757} \log(sales), R^2 = 0.9098, SER = 0.5294.$$

The estimated elasticity between rd and $sales$ is 1.0757. This means that for every 1% increase in sales, a firm spends 1.0757% more dollars in research and development.

```
#The regression object
regression_logrd<-lm(log(rd)~log(sales),data=data)
#Summary without robust errors.
summary(regression_logrd)
#summary with robust errors
coeftest(regression_logrd, vcov = vcovHC(regression_logrd, "HC1"))
```

- (c) Construct a 95% confidence interval for the elasticity of rd with respect to $sales$. **Be sure to use robust standard errors.** From the previous results we can construct a confidence interval using robust errors for the elasticity. Below I propose a method to automatically compute it with R by first estimating the heteroskedasticity robust variance-covariance matrix, then extracting the diagonal elements, which are the estimated variances, and taking their square roots which are the standard errors.

Whether computing them by hand or with the computer, the 95% confidence interval for the elasticity is [0.9634725, 1.187988]. As you can see, 1 is in the confidence interval, so we wouldn't be able to reject a null hypothesis that states that this elasticity is 1, i.e. $H_0 : \gamma_1 = 1$.

```
#Here, we will first extract the standard errors
#from the robust variance-covariance matrix
#estimate. The estimated variances are in the
#diagonal of the matrix. The standard errors,
#are the square root of this diagonal elements.
robustVar_logrd<-diag(vcovHC(regression_logrd, "HC1"))^0.5

#The lower bounds of the confidence intervals are
coef(regression_logrd)-qnorm(1-0.05/2)*robustVar_logrd
#The Upper bounds of the confidence intervals are
coef(regression_logrd)+qnorm(1-0.05/2)*robustVar_logrd
```

4. For this problem, you will use the data file Birthweight_Smoking, which contains data for a random sample of babies born in Pennsylvania in 1989. The data include the baby's birth weight together

with various characteristics of the mother, including whether she smoked during pregnancy. In this problem you will investigate the relationship between birth weight and smoking during pregnancy. For the questions in part (b), **be sure to use robust standard errors**.

Again, we first clean the environment and load the data set. This time we need to use a different command since the data set is in excel.

```
rm(list=ls())
#To clean the environment again

#load the bithweight data set
data<-read_excel("birthweight_smoking.xlsx")
```

(a) In the sample:

- i. What is the average value of *Birthweight* for all mothers?
- ii. For mothers who smoke?
- iii. For mothers who do not smoke?

The average value of *birthweight* is 3382.934 grams. On the other hand, for mothers who smoke, the average *birthweight* is 3178.832 grams and the average *birthweight* for mothers who don't smoke is 3432.832 grams.

Notice that the average difference in *birthweight* between smoking and non smoking mothers is -253.2284 grams.

```
cat("The average birthweight for newborns
for all mothers is", mean(data$birthweight), "\n")

cat("The average birthweight for mothers
who smoke is", mean(subset(data, smoker==1)$birthweight), "\n")

cat("The average birthweight for mothers
who don't smoke is", mean(subset(data, smoker==0)$birthweight), "\n")

cat("The difference in averages between smoking
mothers and non smoking mothers is",
mean(subset(data, smoker==1)$birthweight)
-mean(subset(data, smoker==0)$birthweight), "\n")
```

(b) Run a regression of *Birthweight* on the binary variable *Smoker*. The estimated equation is

$$\widehat{birthweight} = 3432.06 - 253.228smoker, R^2 = 0.0286, SER = 583.7.$$

(11.891) (26.810)

- i. What is the difference in average birth weight for smoking and nonsmoking mothers?
- ii. What is the standard error for the estimated difference in (i)?
- iii. Construct a 95% confidence interval for the effect of smoking on birth weight.

The estimated difference in average birth weight between smoking non smoking mothers is the coefficient on *smoker* which is -253.228 grams, which is precisely what we obtained by computing directly this difference in the previous part. The robust standard error for this difference is 26.810 grams. Finally, the 95% confidence interval for this average difference is [-305.7757, -200.681].

```
#The regression
regression_birthweight<-lm(birthweight~smoker,data=data)
#Summary without robust errors
```

```
summary(regression_birthweight)
#with robust errors
coeftest(regression_birthweight,
  vcov = vcovHC(regression_birthweight, "HC1"))

#Again, we compute as a standalone object the robust standard
#errors
robustVar_birthweight<-diag(vcovHC(regression_birthweight,"HC1"))^0.5

#The lower bounds of the confidence intervals are
coef(regression_birthweight)-qnorm(1-0.05/2)*robustVar_birthweight
#The Upper bounds of the confidence intervals are
coef(regression_birthweight)+qnorm(1-0.05/2)*robustVar_birthweight
```